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A Multi-Objective Optimization-Driven Screening Approach for Maximizing Hydrogen Storage Capacities in MOFs

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Abstract

The inherent tradeoff between gravimetric and volumetric hydrogen storage capacities in metal-organic frameworks (MOFs) limits their commercial viability. While benchmarked MOFs like MOF-5, IRMOF-20, and PCN-610 perform well at 77 K, maintaining their performance at elevated temperatures (298 K) remains challenging. To address this, a multi-objective particle swarm optimization framework was developed to identify promising MOFs for hydrogen storage. The optimization was guided by predictions from the bootstrapped-random forest trees. This optimization yielded 152 theoretical MOF feature combinations, which were matched with 733,792 existing structures. A nearest neighbor search identified 43 promising MOFs, with Zn-based MOF-2087 emerging as the global best, exhibiting **consistent hydrogen storage** performance across temperatures. Grand Canonical Monte Carlo simulations confirmed its high hydrogen uptake (5.3 wt. % and 7.4 gH₂ L⁻¹ at 298 K). Molecular dynamics simulations further revealed C-clusters and metal sites as key adsorption centers, supporting the enhanced hydrogen

storage behavior of MOF-2087. These findings highlight MOF-2087 as a computationally promising MOF for hydrogen storage up to 298 K and demonstrate the effectiveness of the optimization-driven screening strategy.

Keywords: Metal-organic frameworks; Hydrogen storage; Multi-objective optimization; Grand canonical Monte Carlo studies; Molecular Dynamics; Simulation

Introduction

Designing an efficient and reliable hydrogen storage system is crucial for sustainable transportation. Unlike commercially available high-pressure Type-IV cylinders that store hydrogen at 700 bar, metal-organic frameworks (MOFs) offer an effective alternative for storing hydrogen at low pressures due to their exceptional surface areas, tunable properties, and reversible gas storage properties¹⁻⁵. Several factors influence hydrogen storage in MOFs, including storage temperature, pressure, pore geometry, crystal density, heat of adsorption⁶⁻⁹. Identifying commercially viable MOFs with the desired features remains a challenge. Computational screening methods like high-throughput density functional theory¹⁰⁻¹³ and high-throughput grand canonical Monte Carlo (HT- GCMC) simulations¹⁴⁻¹⁷, have been widely used to address this complexity. Due to their computational cost, machine learning (ML) approaches^{12,16-22} have emerged as an efficient alternatives for predicting MOF performance. Supplementary Table 1 summarizes recent ML models applied in this domain. These models can rapidly evaluate the hydrogen storage performance of MOF candidates. However, they require classical benchmark structures like MOF-5 to compare performance and identify best performing candidates.

Ahamed et al. compared the hydrogen storage performance of MOFs from various databases with MOF-5 at 77 K and identified IRMOF-20 and PCN-610 as the best-performing MOFs^{23,24}. Table 1 summarizes the gravimetric and volumetric hydrogen uptake of benchmarked MOFs reported in the literature.

However, they fall short of the U.S. Department of Energy's (DOE) recommended target (6.5 wt. % and 50 gH₂ L⁻¹)²⁵ for the operational temperature range of -40 °C to 60 °C. To address this limitation, researchers have proposed selecting MOFs based on their physical parameters as an effective strategy^{6,21,23,24,26}. Prior studies have shown that hydrogen storage performance in MOFs depends on the combination of crystallographic features, including single crystal density, gravimetric and volumetric surface areas, pore volume, pore diameter, cavity diameters, and void fractions²¹. While hydrogen adsorption studies have traditionally focused on gravimetric performance, works by Goldsmith et al.²⁶ and Ahamed et al.²⁴ have emphasized the need to concurrently consider volumetric uptake at 77 K—keeping the fact that the materials like MOF-5 and HKUST-1²⁷ meet the gravimetric targets but lacks in sufficient volumetric capacity.

Recognizing the limitations of focusing on a single performance metric, Gómez-Gualdrón et al.²⁸ and Chen et al.²⁹ stressed that both gravimetric and volumetric capacities must be concurrently balanced to satisfy the volume and weight constraints of onboard hydrogen storage. However, achieving this balance remains a challenge at higher temperatures. For instance, Chen et al.²⁹ synthesized hypothetical MOFs based on the NU-1500 framework and identified NU-1501-Al as a relatively balanced structure, but it delivered only 2.5 wt.% at 298K. This performance drop can be attributed to the fact that existing studies have largely relied on maximizing the crystallographic features or altering topologies to enhance hydrogen storage³⁰. However, they did not establish the specific combinations crystallographic features required for effective storage. Quantitative benchmarks, such as the optimal range of surface area, pore volume and crystal density were also not defined to **leverage** gravimetric and volumetric hydrogen storage. Without these defined targets, identifying or designing MOFs that can satisfy DOE requirements under real-world conditions remains a challenge.

To overcome the limitations of crystallographic feature-driven studies, researchers have adopted optimization-based approaches to [identify the best performing MOFs](#)³¹. For example, Ghude et al.³² [used](#) Bayesian optimization and evolutionary particle swarm optimization to screen over 98,000 MOF structures, [aiming to identify optimal candidates for hydrogen storage under temperature-pressure swing conditions](#) (100 bar/77 K to 5 bar/160 K).combinations. Singh et al.³³ [used optimization algorithms focused on linker functional groups and identified IRMOF-10 as a best candidate](#) (3.9 wt% and 23 gH₂ L⁻¹ at 77 K). However, these studies primarily relied on single-objective optimization approaches (see **Supplementary Table 2**) and thus, optimal combinations of crystallographic features capable of simultaneously maximizing both gravimetric and volumetric hydrogen storage capacities remain unexplored.

Table 1. Hydrogen storage densities of benchmarked MOFs reported in the literature.

MOF	77 K		298 K	
	Gravimetric capacity (wt. %)	Volumetric capacity (gH ₂ L ⁻¹)	Gravimetric capacity (wt. %)	Volumetric capacity (gH ₂ L ⁻¹)
MOF-5 ³⁴⁻³⁸	4.5	31.1	0.8	8.8
IRMOF-20 ³⁴⁻³⁸	5.7	33.4	1.45	9.42
PCN-610 ²³	10.1	35.5	2.66	9.17

Thus, a multi-objective optimization-based screening framework is proposed in this work to identify the MOFs having the optimal crystallographic features for achieving the desired [trade-off](#) between gravimetric and volumetric capacities (Fig. 1). A bootstrap aggregate random forest tree (B-RFT) technique embedded with multi-objective particle swarm optimization (MOPSO)³⁹⁻⁴¹ was used to screen the MOFs from nearly a million MOF structures reported in various databases. Among those, 152 feasible combinations of MOF features were identified [that represent optimal trade-](#)

offs between hydrogen storage densities. Using the Nearest Neighbor Search (NNS) algorithm, MOF-2087 was identified as the top-best performing MOF based on optimized crystallographic feature similarity. Hereafter the term top-best performing candidate is referred as 'global best MOF' for ease of reference. GCMC simulations confirmed that MOF-2087 demonstrates the desired trade-off between gravimetric and volumetric hydrogen storage capacities, surpassing the classical benchmark MOFs at 298 K. Furthermore, molecular dynamics (MD) simulations indicate that C₂, C_R, and Zn atoms in MOF-2087 are the critical sites for hydrogen adsorption and are responsible for achieving a better tradeoff between gravimetric and volumetric storage capacities. Although feature optimization was performed at 77 K due to the scarcity of reliable adsorption data at high temperatures, the optimum features are structural and remains stable up to 298 K. GCMC simulations across 77-298 K showed consistent hydrogen storage performance for MOF-2087, and MD simulations confirmed its structural stability. This supports that optimization at 77 K can reliably identify MOFs performing well at room temperature. Therefore, our results propose MOF-2087 as the new benchmark material for hydrogen storage applications.

Fig. 1. Steps involved in the B-RFT-MOPSO-based deliverable gravimetric and volumetric hydrogen storage optimization of MOFs. (1) Before all else, the realm of 925,229 MOF structures was collected from HyMARC. (2) Formation and parameter tuning of B-RFT framework to predict the hydrogen storage densities in MOFs, later the multi-objective framework has been designed by embedding the B-RFT framework with MOPSO. (3) The NNS was applied to match the obtained Pareto-optimal solutions with the existing structures. Further, TOPSIS identifies the global best one. (4) The hydrogen storage performance and features of the global best MOF will be demonstrated by using GCMC studies. (5) The hydrogen storage mechanism of the global best MOF will be evaluated by using MD simulations.

2 Results

To implement the proposed optimization-based screening strategy, the workflow began with collecting 733,792 open-source MOFs from the HyMARC data hub (Fig. 1). This resource complies crystallographic

information from 19 databases worldwide (see Methodology Section 4.1). This collection is hereafter referred to as ‘global database’. Using this dataset, the B-RFT—model was trained to predict gravimetric and volumetric hydrogen storage capacities based on crystallographic features. Following hyperparameter tuning and validation, the model was integrated into MOPSO framework to identify optimal design features. The global best MOF was further evaluated through GCMC and MD simulations. The subsequent sections detail the model development, optimization process, and performance evaluation.

2.1 Hyperparameter tuning of B-RFT Framework

The bootstrap-random forest tree (B-RFT) model combines bootstrap bagging with decision-tree ensembles, where each tree is trained on a random subset of the data. Bagging inherently provides an unbiased internal cross-validation via Out-Of-Bag (OOB) samples, allowing direct estimation of prediction error without a separate hold-out set. An OOB error-based sensitivity was performed to optimize the number of trees (nTrees) and number of leaves per tree (nLeaves) on a representative training subset. Although OOB cross-validation sufficed, a separate validation using 20% of the dataset was maintained throughout to confirm the model’s predictive performance. The nTrees were varied from 1 to 500 to balance computational efficiency with prediction accuracy (**Supplementary Fig. 1**). For gravimetric capacity; the OOB mean square error stabilized at 0.037 after 30 trees, whereas for volumetric capacity, the minimum error of 1.97 was observed at 48 trees, leading to the choice of nTrees as 50. The higher OOB error in volumetric capacity prediction is likely due to the marginal contributions of pore volume, large cavity and pore limiting diameters of MOFs (**Supplementary Fig. 2**). With nTrees fixed at 50, varying the nLeaves between 1 and 100 showed that the OOB error reached a minimum of 0.033 for gravimetric and 1.97 for volumetric capacities with five leaves. Increasing the nLeaves beyond five added model complexity and increased computation time without

significantly improving the prediction accuracy^{42,43}. Therefore, the final configuration of nTrees=50 and nLeaves =5 was selected considering the balance between computational efficiency and prediction accuracy.

2.1.1 Dataset size variation

Beyond nLeaves and nTrees, the quality and quantity of the training dataset significantly affect the prediction accuracy of the B-RFT model^{21,44}. However, determining the exact amount of data required to train a B-RFT model for hydrogen storage prediction with a coefficient of determination (R^2) >0.95 remains unclear. To address this, a sensitivity analysis was performed by varying the dataset size from 5% to 80% (up to 78,956 MOFs), with the model tested on 9,870 MOFs. Fig. 2 (A and B) illustrates the prediction performance of the B-RFT framework for gravimetric hydrogen storage. As shown in Fig. 2 (C and D), when less than 45% of the dataset was used, the performance metric R^2 was substantially lower, while the average unsigned error (AUE), root mean square error (RMSE), and median average error (MAE) were notably higher, especially for volumetric capacity predictions. This decrease in accuracy is likely due to crystallographic inaccuracies and the inclusion of MOFs with zero surface area in the training dataset²¹ (Supplementary Fig. 2). Increasing the size of the training dataset beyond 45% consistently improved the AUE and R^2 metrics. The best balance between computational cost and prediction accuracy was achieved with 75% of the dataset (74,022 MOFs) and was selected for training the optimal B-RFT model. This optimized B-RFT model was subsequently used as the objective function in the MOPSO framework (Fig. 1). Table 2 summarizes the predictive performance of the optimized B-RFT model.

Table 2. Prediction performance of B-RFT framework under pressure swing (100 bar to 5 bar) condition at 77K

Parameter	R^2	AUE	RMSE	MAE
Gravimetric capacity	0.997	0.01 wt.%	0.163 wt.%	0.012 wt.%

Volumetric capacity	0.985	0.209 gH ₂ L ⁻¹	1.37 gH ₂ L ⁻¹	0.056 gH ₂ L ⁻¹
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Fig. 2. The performance assessment of the B-RFT framework concerning the training dataset size is tested by using 9870 MOFs uniformly for all training dataset combinations.

(A) and (C) variation of R^2 (navy blue) and AUE (green) concerning the training dataset size for both the UG and UV uptakes. (B) and (D) RMSE (orange) and MAE analysis (navy blue) of the trained B-RFT framework concerning the training dataset size utilized for UG and UV uptakes.

2.1.2 Trends and correlation between variables

Fig. 3 presents an analysis of the crystallographic feature trends for predicted hydrogen storage capacities across MOFs in the global database, using the best-performing B-RFT model. The analysis reveals that an increase in gravimetric surface area, pore volume, void fraction, pore limiting diameter, and large cavity diameter enhances volumetric and gravimetric hydrogen uptakes. In contrast, single crystal density and volumetric surface

area correlate negatively with both capacities, suggesting that minimizing these features leads to higher hydrogen storage densities. These trends provide a base for defining bounds for crystallographic features utilized in the B-RFT-MOPSO optimization process.

2.1.3 Constraints for hydrogen storage optimization in MOFs

The global MOF database is composed predominantly of hypothetical MOFs (hMOFs) - 97.3% of 733,792 - characterized by low single crystal densities ($< 0.3 \text{ g cm}^{-3}$), high surface areas ($> 5300 \text{ m}^2 \text{ g}^{-1}$), and pore volume exceeding $3.3 \text{ cm}^3 \text{ g}^{-1}$. In contrast, experimental MOFs (eMOFs) tend to have void fractions below 0.87, with only a few exceeding this threshold (refer to Fig. 3). When considering feature combinations such as density $< 0.3 \text{ g cm}^{-3}$, surface area $> 5300 \text{ m}^2 \text{ g}^{-1}$, pore volume $> 3.3 \text{ cm}^3 \text{ g}^{-1}$, and void fraction ≥ 0.87 , no MOFs in the global database satisfy all simultaneously. Although Fig 3G-H suggests that the optimal void fraction range lies between 0.87 and 0.92, only MOFs within the narrower range of 0.89-0.91 meet all three other criteria concurrently. This observation is consistent with recent computational and experimental studies^{29,31}. Therefore, the void fraction range in this work has been refined to 0.89-0.91 for effective screening.

To eliminate non-viable solutions for MOF crystallographic features, specific parametric constraints were incorporated into the B-RFT-MOPSO framework (**Supplementary Note 3**). Based on the feature trend patterns, it is evident that MOFs with low densities and high surface areas are particularly conducive to achieving better gravimetric and volumetric densities, especially under pressure swing conditions (100 bar - 5 bar). Considering the availability of MOFs and feature correlations, maximum and minimum bounds for B-RFT-MOPSO have been established, as detailed in **Supplementary Table 3**. **Supplementary Fig. 3** provides a two-dimensional visualization of the search domain designed to identify optimal MOFs with enhanced hydrogen storage capacities.

Fig.3. Correlation and trends of seven crystallographic features of MOFs concerning UG and UV. Variation of deliverable capacities of global database concerning the single crystal density, surface areas, void fraction, accessible pore volume, cavity diameters, and pore limiting diameters of MOFs assuming the pressure swing conditions (100-5 bar) at 77K. (A, C, E, G, I, K, M) represents the UGs of MOFs and (B, D, F, H, J, L, N) represents the UVs of MOFs.

2.2 Performance evaluation of B-RFT-MOPSO

A series of performance assessment indicators were utilized to evaluate the diversity and convergence of the B-RFT-MOPSO optimization framework (**Supplementary Method 3**). As the true Pareto front of the multi-objective hydrogen uptake problem is unknown, a reference Pareto front was constructed using 152 feasible points obtained from 15 independent runs of B-RFT-MOPSO, serving as a baseline for evaluation (**Supplementary Fig. 4, red markers**). Generally, the size of the obtained Pareto set reflects the ability of the B-RFT-MOPSO to identify feasible non-dominated solutions. On average, approximately 47 feasible non-dominated solutions were generated per run, consistent with optimization models reported in the literature⁴⁵ (**Supplementary Fig. 5(A)**).

Diversity in the obtained Pareto-optimal solutions was evaluated using the standard spacing metric (SSM) indicator, where a lower SSM value signifies better diversity in the predicted non-dominated solutions. The mean SSM value of 0.0002 indicates that the B-RFT-MOPSO consistently achieves well-diverged non-dominated solutions across independent runs (**Supplementary Fig. 5(B)**). The range of SSM values observed aligns with the existing studies⁴⁶.

The quality of the Pareto points generated by the B-RFT-MOPSO was assessed by their proximity to the reference or true solution sets, measured by the standard generational distance (SGD) indicator (**Supplementary Method 3.3**). A smaller SGD value indicates that the non-dominated solutions obtained in each run are closer to the reference Pareto points. The mean SGD value of 0.0235 for the B-RFT-MOPSO framework demonstrates its ability to effectively identify MOF features that enhance hydrogen storage (refer to **Supplementary Fig. 5(C)**).

Additionally, the search domain of the B-RFT-MOPSO in the global search space is significantly influenced by the convergence rate of the framework.

Forcing a multi-objective model to achieve rapid convergence can limit the search space exploration. Therefore, the total number of framework evaluations required to reach the desired Pareto points was used to assess the convergence rate of the designed multi-objective framework⁴⁷. On average, 109,437 simulations were performed in each independent B-RFT-MOPSO run to obtain feasible non-dominated solutions.

2.3 Feasible crystallographic features of MOFs and their usable hydrogen storage capacities

2.3.1 Gravimetric Hydrogen Storage in MOFs

The B-RFT-MOPSO optimization framework identified 152 non-dominated crystallographic features of MOFs (**Supplementary Fig. 4 - navy blue**) that exhibit a feasible tradeoff between hydrogen storage capacities under pressure swing conditions at 77 K. The resulting non-dominated solution set indicates that reducing single-crystal density (0.24 g cm^{-3} to 0.10 g cm^{-3}) and volumetric surface area ($1800 \text{ m}^2 \text{ cm}^{-3}$ to $500 \text{ m}^2 \text{ cm}^{-3}$) enhances gravimetric hydrogen uptake in MOFs (**Supplementary Fig. 6(A and B)**). In contrast, maximizing gravimetric surface area ($\sim 7200 \text{ m}^2 \text{ g}^{-1}$) results in improved hydrogen storage, aligning with Chahine's rule²⁹ (**Supplementary Fig. 6(C)**). Void fractions from 0.905 to 0.91 also favor gravimetric hydrogen storage. Additionally, an increase in accessible pore volume correlates with enhanced gravimetric uptake. The highest gravimetric hydrogen storage was observed in MOFs with large cavity diameters between 16 Å and 18 Å and pore-limiting diameters between 14 Å and 18 Å (**Supplementary Fig. 6(F and G)**).

2.3.2 Volumetric Hydrogen Storage in MOFs

In contrast to gravimetric hydrogen storage, an increase in single-crystal density enhances volumetric hydrogen storage (**Supplementary Fig. 6(A)**). A similar trend is observed for MOFs' volumetric and gravimetric surface areas. Void fractions in the range of 0.905 to 0.91 result in higher volumetric

hydrogen storage, with a maximum of approximately $38 \text{ gH}_2 \text{ L}^{-1}$ (as the dataset is capped below $40 \text{ gH}_2 \text{ L}^{-1}$)²¹ for large cavity and pore-limiting diameters between $16 \text{ \AA}$ and $18 \text{ \AA}$, and $14 \text{ \AA}$ and $18 \text{ \AA}$, respectively.

The identified non-dominated solutions represent feasible combinations of crucial crystallographic features in MOFs and serve as theoretical benchmarks for optimizing hydrogen storage tradeoffs. However, it is challenging to computationally construct or reconstruct new MOF structures based on these crystallographic features. To overcome this, existing MOFs from the global database were matched with the optimal features using the nearest neighbor search (NNS) algorithm. If a MOF's features closely align with the optimal solution, it is identified as a locally best MOF and stored in a local repository.

2.4 Local Best MOFs

Applying the nearest neighbor search (NNS)^{48,49} to the global database of 733,792 MOFs, the crystallographic features of each MOF were compared to the query points (optimal solutions found from B-RFT-MOPSO) to identify the local best MOFs. The degree of similarity between the features of each MOF and query points was determined using a customized 7-dimensional Euclidean distance (ΔEx) metric. MOFs with the smallest ΔEx for a specific query point in the global repository were identified as the local best MOF and transferred to the local-best repository. In total, 43 out of 733,792 MOFs were identified as the closest match to the 152 Pareto optimal solutions. These identified local best MOFs are illustrated in **supplementary Fig. 4**. Notably, all the screened local best MOFs were hypothetical.

Among the various MOF databases, the TobaCCo and UO databases demonstrated the most feasible tradeoffs between hydrogen storage densities under pressure swing conditions at 77 K. However, 10 MOFs were identified as significantly distant from their respective query points. To refine

the selection, a local domain was implemented to narrow down further MOFs that were sufficiently close to the optimal solutions (**Supplementary Fig. 4**). Fig. 4 presents the shortlisted local best MOFs and the corresponding Pareto optimal solutions obtained through the B-RFT-MOPSO framework.

Fig. 4. Pareto optimal solutions obtained from B-RFT-MOPSO. The orange-coloured points indicate the shortlisted local best MOFs and their respective closeness to the Pareto solutions. The greenish point represents the global best MOF feature identified by the Technique for Order of Preference by Similarity to the Ideal Solution (TOPSIS).

2.5 Global best MOF

Each crystallographic feature combination derived from the B-RFT-MOPSO framework represents a unique tradeoff between gravimetric and volumetric hydrogen uptake. Thus, identifying the single best solution that demonstrates a feasible [trade-off](#) between hydrogen storage capacities in MOFs is essential. To determine the global best solution, the TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) multi-criteria decision-making method was employed⁵⁰ (**Supplementary Method 3.4**). **Supplementary Table 4** presents the top 5 hMOFs ranked by the closeness

index. The feature ranked first exhibits the most feasible tradeoff between hydrogen storage capacities, making it an optimal solution suitable for inclusion in the decision-maker's repository.

Using the identified global best solution as a benchmark for the best-performing MOF, the NNS identified MOF-2087 from the ToBaCCo database²² as the optimal MOF with the minimal ΔEx value of 17.8, surpassing the other MOFs in the global database. Table 3 provides the crystallographic features of MOF-2087, along with the single-objective dominant solutions and their respective hydrogen storage capacities. The MOF from the UO database²⁰ exhibits high volumetric storage of 37.85 gH₂ L⁻¹ (Str_m3_o7_o19_f0_nbo.sym.22.out, a volumetric dominant solution) with a moderate gravimetric capacity of 12.2 wt.% under pressure conditions at 77 K. Conversely, MOF-12358 from the ToBaCCo database was identified as the gravimetric storage-dominant solution, achieving 21.86 wt. % and 31.84 gH₂ L⁻¹ among the local best MOFs. As listed in Table 3, the crystallographic features of MOF-2087 align with the theoretical benchmark features.

Table 3. A comparison between optimized solutions and best-performing MOFs.

Criteria	Density (cm ⁻³)	GSA (m ² g ⁻¹)	VSA (m ² cm ⁻³)	VF	PV (cm ³ g ⁻¹)	LCD (Å)	PLD (Å)	Predicted UG (wt.%)	Predicted UV (gH ₂ L ⁻¹)
Gravimetric capacity dominated solution	0.1	5573.97	557.39	0.90	9.09	19.41	17.27	21.27	34.63

MOF-12538*	0.107	5544.2	598.40	0.95	8.87	59.35	43.61	21.86	31.85
Feasible MOF identified by TOPSIS (Rank-1)	0.133	7200	958.55	0.90	6.82	17.75	14.54	17.95	35.7
MOF 2087*	0.132	7188.7	949.68	0.93	7.08	26.03	21.15	19.16	34.45
Volumetric capacity dominated Solution	0.247	7198.36	1778.78	0.89	3.63	18.03	10.94	12.18	37.94
Str_m3_o7_o19_f0_nbo.sym.22.out*	0.24	7174	1744	0.89	3.68	18	10.7	12.20	37.85

* MOF from global database

2.6 Hydrogen storage performance of global best MOF: MOF-2087

2.6.1 Structural descriptions of MOF-2087

MOF-2087 framework consists of a zinc (II) metal cluster bonded to the nitrogen atom in the pyridine ring of the organic ligand (3-ethynyl-7-(pyridine-4yl) dibenzofuran), forming a 3D structure with a dia net topology (4/6/c1; sqc6), as shown in Fig. 5. The ethynyl terminals of the four ligand units under cyclo-addition, resulting in the formation of an 8-membered C-cluster (bicycle-[2,2,2]-octane) in a tetrahedral arrangement (Fig. 5(C)). Pore size distribution analysis of MOF-2087 reveals two adjacent peaks at 23.5 Å and 23.9 Å, indicating the mesoporous nature of the material (**Supplementary Fig. 7**). Furthermore, the crystallographic information of MOF-2087 can be found in the **Supplementary document**.

Fig. 5. The pictorial representation of MOF-2087. (A) 2x2x2 supercell of MOF-2087 facing towards the hexagonal pore geometry. (B) Organic-ligand. (C) 8 membered C-cluster. (D) hexagonal pore geometry. C-grey, H-white, N-blue, O-red, and Zn-violet.

2.6.2 Hydrogen storage mechanism

2.6.2.1 Preferential sites of hydrogen interaction

The hydrogen density distribution and radial distribution functions (RDF) at 77 K and 5 bars were calculated to determine the preferential hydrogen adsorption sites in MOF-2087 (Fig. 6). The strongest hydrogen interactions were observed with C atoms at 3.8 Å, followed by O at 4.0 Å, and H at 4.04 Å. Additionally, N-H₂ interactions peaked at 4.8 and Zn-H₂ interactions were observed at 5.31 Å. These results suggest that organic elements such as C, O, and H exhibit stronger interactions with H₂ molecules than the Zn metal center. The observed results were consistent with the results from literature^{51,52}.

The supercell (2x2x2) of MOF-2087 was meshed into 500x500x500 grids to compute Gaussian and normalized H₂ density distributions, visualized using ParaView®. Fig. 7 (A, B and C) represents the face and axillary projections of MOF-2087 before adsorption, while Fig. 7 (D, E and F) illustrates MOF-2087 after hydrogen adsorption at 50 bar and 77 K. Correlating the RDF with the mass center probability, significant hydrogen interactions were observed around the 8-membered C-clusters with tetrahedrally coordinated dibenzofuran rings, forming an irregular, diagonally stretched square-shaped adsorption site (Fig. 6(A), labeled as Site I). Additional interactions were identified at the Zn sites (Fig. 6(B and C), labeled as Site II) and around the dibenzofuran rings (labeled as Site III). Hydrogen density mapping reveals comprehensive H₂ molecule distribution within the hexagonal pore, spanning the (011) and (110) crystallographic planes (Fig. 7(E and F), and **Supplementary Fig. 8**). This distribution highlights trapping of hydrogen within the pores surrounded by dibenzofuran rings, Zn sites, and 8-membered C-clusters, contributing to a high hydrogen uptake of 22.3 wt. % and ~38 gH₂ L⁻¹ at 77 K and 100 bar. These findings indicate that hydrogen adsorption sites in MOF-2087 are primarily located in the organic ligands (Site I and III), rather than the metal site (Site II)^{51,52}.

2.6.2.2 Simulated Annealing

To further understand the hydrogen storage mechanism of MOF-2087 and validate the GCMC findings, classical MD simulations were conducted. C atoms in the ligand were classified as C_2 (from Site I and III), C_3 (from Site I), and C_R (from Site III), based on their geometry and force field classifications⁵³ (Fig. 8). Preliminary GCMC simulations revealed that a MOF-2087 supercell can adsorb approximately 188 H₂ molecules at 77 K and 1 bar; therefore, 188 H₂ molecules were introduced into the simulation cell for the MD simulations⁵⁴. Fig. 9 displays the annealed structure of the MOF-2087-H₂ system within the NVT ensemble. The thermal motion and equilibrium zones of H₂ were characterized by measuring the peak RDF distances between host

atoms and H₂ (Table 4). These distances indicate that, as the system cools, H₂ molecules tend to move closer to specific atoms, suggesting favorable adsorption interactions. This behavior is likely due to reduced thermal motion at lower temperatures, allowing H₂ molecules to settle into lower-energy adsorption sites within the framework⁵⁵.

Table 4 shows a significant decrease in the average distance between C_2 and H₂ from 5.9 Å at 300 K to 3.6 Å at 14 K, indicating that these atoms become favorable adsorption sites, likely due to π - π interactions resulting from resonance in the rings. Similarly, the distance between C_R and H₂ decreases from 6.1 Å to 3.6 Å, reflecting stronger π - π stacking interactions. As the temperature increases, the distance between Zn and H₂ increases. However, as the system cools, this distance shortens from 8.3 Å to 5.0 Å. This suggests that while Zn centers are not the main adsorption sites, they may still participate in adsorption at lower temperature or under high pressure, possibly due to polarization effects. O atoms also exhibit strong interactions, with the distance between O and H₂ decreasing from 6.2 Å to 4.2 Å, indicating an affinity for H₂ molecules at lower temperatures. Additionally, the distance between host H atoms and H₂ decreases from 6.8 Å to 3.9 Å, indicating preferential adsorption. These H atoms belonging to the pyridine and furan moieties⁵⁴, are accounted for alongside the C_R atoms throughout the study. Overall, the annealed structure demonstrates that C_R and C_2 carbons are the most favorable adsorption sites for H₂, primarily due to the electronegativity of the sp² hybridized carbon atoms, followed by O atoms. These findings support the preliminary adsorption sites identified in the GCMC simulations (Fig. 6).

Table 4. Calculated distance of peak $g(r)$ of host - H₂ molecules interactions concerning the cooling temperature.

Temperature (K)		Radial distance (Å)						
Initial	Final	C_2	C_3	C_R	N	Zn	O	H
300	250	5.9	5.8	6.1	7.2	8.3	5.2	6.8
250	200	7.3	7.4	6.2	7.0	5.7	6.4	7.2
200	150	7.0	4.8	6.7	8.3	6.2	4.8	6.9
150	100	5.9	5.4	4.4	5.3	5.6	4.5	4.5
100	50	3.8	4.1	3.9	6.3	5.5	6.2	4.1
50	14	3.6	4.3	3.6	4.7	5.0	4.2	3.9

Fig. 6. Mass center probability of hydrogen distribution in MOF-2087 (red-x axis, green-y axis, and blue-z axis) (A, B, and C) respectively. (D) represents the radial distribution function of gaseous hydrogen concerning the radial distances of the elements presented in MOF-2087, evaluated at 77K and 5 bar.

Fig. 7. Pictorial representation of the empty MOF-2087 supercell (A, B, and C) and hydrogen density distribution of the MOF-2087 under 77 K and 5 bar (D, E, and F) seen through the respective crystallographic planes (red-x axis, green-y axis and blue-z axis).

Fig. 8. Pictorial representation of classified carbon atoms according to the DERIDING force field.

Fig. 9. Annealed structure of MOF-2087 at different projections (A and B).

The H₂ molecules found in the first interaction zone ($g(r)$) are mapped in orange, magenta, green, yellow, and red for O, C_2, C_R, N, and Zn, respectively. For visualization purposes, the H₂ molecules are scaled larger than their actual size.

2.6.2.3 H₂ diffusion in MOF-2087

Mean square displacement (MSD) statistics were collected from equilibrated conditions throughout 2000 ps. The calculated MSD for all temperatures considered (77 K, 160 K, 200K, and 298 K) exhibits a linear trend, making it suitable for computing H₂ diffusivities within the MOF, as shown in Fig. 10. The diffusion characteristics of H₂ inside MOF-2087 demonstrate a clear trend of increasing diffusion rates with rising temperature, from $3.78 \times 10^{-8} \text{ m}^2 \text{ s}^{-1}$ at 77 K to $2.92 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$ at 298 K. This trend indicates enhanced H₂ mobility with increasing thermal energy, enabling H₂ molecules to overcome potential energy barriers within the MOF and explore the structure more freely⁵⁶.

This behavior aligns with the simulated annealing results, where reduced thermal energy at lower temperatures leads to minimal diffusion rates, allowing H₂ molecules to access and occupy favorable adsorption sites. As illustrated in Fig. 10(B), the diffusion coefficients of MOF-2087 exhibit

Arrhenius behavior ($R^2 = 0.98$) across the studied temperature range. The calculated activation energy of 0.017 eV is lower than that of MOFs without open metal sites, such as Al-soc-MOF-1d (0.034 eV)⁵⁶, IRMOF-1 (0.026 eV), IRMOF-8 (0.0222 eV), and IRMOF-18 (0.032 eV)⁵⁷, and slightly higher than that of MOF-5 (0.014 eV)⁵⁶. This moderate activation energy, combined with the lower self-diffusion coefficient of MOF-2087, suggests that the strong interaction between H_2 and host adsorption sites tends to localize H_2 molecules around organic nodes, particularly carbon clusters, resulting in reduced diffusivity.

Fig. 10. MSD of H_2 at various temperatures (A) and Arrhenius plot of MOF-2087 ($R^2 = 0.98$) (B).

2.6.2.4 H_2 displacement in MOF-2087

To gain a deeper understanding of H_2 displacement within MOF-2087, radial distribution functions of H_2 molecules relative to host atoms were computed across different temperature zones (Fig. 11 and 12). A sharp peak at 3.65 Å for C₂ and Zn indicates that many H_2 molecules preferentially settle at these sites. At ambient temperatures, the H_2 molecules are more dispersed and accumulate around 4.65 Å and 8.46 Å near Zn, likely due to increased thermal motion at higher temperatures. Significant H_2 interactions were observed at

4.4 Å for C₃ at 77 K, reflecting the low-temperature effects noted during simulated annealing. However, this high-intensity peak for C₃ becomes saturated and shifts to 4.53 Å and 6.53 Å at 298 K, indicating that, similar to Zn, C₃ effectively participates in H₂ adsorption at lower temperatures.

The O–H₂ interaction peaks were observed at 3.71 Å at 77 K. While a shoulder peak is observed at 2.90 Å, inferring O becomes a more favorable adsorption site at 298 K. The number of H₂ molecules distributed at 77 K and 298 K provides insights into the nature of hydrogen interactions with specific host atoms at varying distances (Table 5). The average number of H₂ molecules around C₂ and C_R atoms is significantly higher at 77 K, especially within the 5 Å to 6 Å range. Although strong interactions persist at ambient temperature, the number of H₂ molecules decreases due to the increased thermal motion. This suggests that C₂ and C_R are more favorable adsorption sites than others.

As a metal site, Zn exhibits moderate interaction with hydrogen at 77 K and 298 K, indicating that while Zn is not the primary adsorption site, it likely plays a supportive role in stabilizing hydrogen molecules. At 77 K, the H₂ distribution number around O atoms is 0.48 at a 6 Å distance, underscoring O's significant role in H₂ adsorption. However, this value decreases at 298 K, showing that higher temperatures reduce the likelihood of sustained H₂ adsorption at O sites. It is important to note that comparing these numbers with the enhanced H₂ uptake of MOF-2087 might be misleading, as the number of atoms in the supercell is relatively high; for instance, MOF-2087 contains 820 C₂ atoms within its supercell. Therefore, the collective contribution of individual host atoms results in substantial H₂ adsorption, exceeding that of any single adsorption site. Correlating the MD results with GCMC findings confirms that C₂ and C_R (from sites I and III) are the primary adsorption sites in MOF-2087. While O and Zn atoms also contribute to adsorption, their effectiveness is more temperature-dependent.

Fig. 11. Computed radial distribution function of host-H₂ interaction at 77 K. (A) classified carbon atoms. (B) host atoms from site II and III.

Fig. 12. Computed radial distribution function of host-H₂ interaction at 298 K. (A) classified carbon atoms. (B) host atoms from site II and III.

Table 5. Calculated number of H₂ molecules found with respect to the radial distance of host atoms.

Distance (Å)	77 K						298 K						
	C_2	C_3	C_R	N	Zn	O	C_2	O	C_2	C_3	C_R	N	Zn

4	0.15	0.058	0.11	0.11	0.06	0.096	0.04	0.09	0.06	-	0.06	0.009	-	
5	0.32	0.43	0.3	0.28	0.23	0.22	0.16	0.26	0.15	0.02	0.2	0.1	0.03	0
6	0.53	0.67	0.51	0.52	0.64	0.48	0.35	0.42	0.31	0.12	0.31	0.3	0.2	

In addition to the structural parameters and preferential adsorption sites, the relative binding characteristics of MOFs contribute to hydrogen storage capacities. The estimated isosteric heat of hydrogen (Q_{st}) of MOF-2087 is 3.1 kJ mol⁻¹, indicating physisorption based hydrogen storage rather than chemisorption (**Supplementary Fig.9**). It is also noteworthy that the changes in pressure or hydrogen loading has shown minimal variations in Q_{st} , with a maximum observed value of 3.18 kJ mol⁻¹ at 100 bar (298 K). This lower Q_{st} suggests that MOF exhibits modest guest-host interactions. While the substantial H₂ storage capacities observed in MOF 2087 is driven by sites I and II and the framework's considerable porosity²⁹.

2.6.3 Hydrogen storage performance of MOF 2087

The hydrogen sorption characteristics of MOF-2087 were calculated using the RASPA code⁵⁸. Fig. 13 illustrates the estimated total gravimetric and volumetric H₂ uptake of MOF-2087 across varying pressures (0-100 bar) and temperatures (77 K to 298 K). The hydrogen adsorption isotherm at 77 K (Fig. 13) shows a steep increase in H₂ uptake at low-pressure regimes (< 20 bar), indicating a physisorption-based hydrogen storage mechanism⁵⁹. The uptake reaches a maximum of approximately 38 g H₂ L⁻¹ and 22.3 wt. % at 100 bar. Despite MOF-2087 exhibiting a relatively low isosteric heat of adsorption, which influences low-pressure hydrogen storage, its substantial free volume, surface area, and adsorption sites contribute to enhanced H₂ uptake at medium and high-pressure regimes, achieving an optimal [trade-off](#) between hydrogen storage capacities⁶.

The H₂ adsorption isotherm of MOF-2087 demonstrates a monotonic increase in uptake with increasing pressure and does not saturate at 100 bar, indicating structural stability under high-pressure conditions. Additionally, hydrogen density maps at low, medium, and high pressures show active H₂ interactions primarily at the site I, with contributions from sites II and III during higher loadings (**Supplementary Fig. 8**). The observed linearity in the isotherm at elevated temperatures (160 K to 298 K) is attributed to the dilution of the adsorbed H₂ layer on the surface of the MOF, a behavior similar to that reported by Assoulaye et al.³⁸.

The hydrogen storage capacities of MOF-2087 were evaluated at operating temperatures ranging from 77 K to 298 K, as shown in Fig. 13(C). Under pressure swing operations, MOF-2087 achieves a maximum hydrogen storage capacity of 32.4 gH₂ L⁻¹ and 18.3 wt. % at 77 K, in alignment with the B-RFT-MOPSO predictions (Table 3 and Fig. 13(C)). To the best of our knowledge, MOF-2087 demonstrates a more **trade-off** hydrogen storage density compared to classical MOFs available within the specified search domain. As expected, MOF-2087 retained this trade-off at elevated temperatures, as shown in Fig. 13(C). A reduction in less than 10 % in both gravimetric and volumetric capacities was observed between 200 K and 298 K, indicating the MOF's stability in maintaining enhanced hydrogen uptake at DOE-recommended operating temperatures. Specifically, at ambient temperature, MOF-2087 achieves a maximum uptake of 5.3 wt. % and 7.4 gH₂ L⁻¹, indicating an **optimized** gravimetric and volumetric hydrogen storage performance.

Fig. 13. Hydrogen adsorption characteristics.

(A) Total gravimetric and (B) volumetric hydrogen adsorption of MOF-2087.

(C) UG and UV of MOF 2087 under pressure swing conditions (100 bar - 5 bar) concerning the temperature.

3. Discussion

The primary objective of the present study is to identify a MOF having a superior tradeoff between gravimetric and volumetric hydrogen capacities under pressure swing conditions at 77 K. By coupling the prediction accuracy of the B-RFT model with the search efficiency of MOPSO, the optimization-based screening approach effectively optimized the crystallographic features of MOFs and identified a better solution from the vast global database. The

hyperparameter tuning and optimized training dataset size for the B-RFT model significantly improved prediction accuracy (**Supplementary Table 1**). Additionally, single-crystal density was found as a key feature, negatively correlating with both hydrogen storage capacities, unlike other features^{21,23}. Specific sweet spots were observed for volumetric hydrogen storage in terms of single crystal density ($\sim 0.5 \text{ g cm}^{-3}$)²³, the void fraction (~ 0.90), pore volume ($\sim 5 \text{ cm}^3 \text{ g}^{-1}$), large cavity diameter ($\sim 15 \text{ \AA}$), and pore limiting diameter ($\sim 8 \text{ \AA}$) of MOFs (Refer to Fig. 3). The B-RFT-MOPSO framework successfully maintained these sweet spots (**Supplementary Fig. 6(F and G)**), underscoring the accuracy and search efficiency of the approach. While these features exhibit their respective sweet spots, achieving the optimal combination remains hypothetical; hence, a $7 \text{ \AA} - 9 \text{ \AA}$ difference between the rank one feature and cavity and pore limiting diameters of MOF-2087 was observed.

Sorting the features in the HyMARC dataset based on the design space constraints reveals a volumetric ceiling of $40 \text{ gH}_2 \text{ L}^{-1}$, indicating that no MOF in the dataset meets the DOE's volumetric targets (**Supplementary Fig.10**). For example, the H74A database contains MOFs with the gravimetric surface areas between $3226\text{--}3704 \text{ m}^2 \text{ g}^{-1}$, below the desired range of $5300\text{--}7200 \text{ m}^2 \text{ g}^{-1}$. Meanwhile, databases like UO, SNW, and ToBaCCo show promising numbers of MOFs (5169, 346, and 794, respectively) with favorable gravimetric-volumetric tradeoffs. However, some entries in CSM-2018-I have densities ($0.33\text{--}3.22 \text{ g cm}^{-3}$) and void fractions in MTV (beyond $0.89\text{--}0.91$) that fall outside the design range. Analyzing the non-dominated crystallographic features of MOFs (**Supplementary Fig.6**), a reduction in single crystal density and volumetric surface area (refer to **section 2.3**) enhances gravimetric hydrogen uptake. This should be prioritized in designing MOFs for hydrogen storage applications, where maximizing energy density per unit weight is crucial, especially for automotive applications. In contrast, high-density MOFs with void fractions between 0.905 and 0.91 , large cavity

diameters (16-18 Å), and pore-limiting diameters (14-18 Å) provide an optimal [trade-off](#) for achieving storage capacities approaching DOE targets.

The B-RFT-MOPSO framework was restricted to a defined search domain, but additional best-performing MOFs may exist outside this space.

One of the significant challenges to consider here is synthesizing these hypothetical MOFs and achieving the desired levels of porosity and surface area^{21,23,24}. Despite this, observations using B-RFT-MOPSO combined with NNS and TOPSIS identified the Zn-based MOF-2087 as the global best-performing MOF in terms of optimal crystallographic feature combinations. Previous studies demonstrated the classical benchmark MOFs such as MOF-5, IRMOF-20, and PCN-610, are capable of having better tradeoffs at 77 K^{21,23,26}. However, these MOFs perform poorly at ambient temperatures, achieving volumetric density of 8.8-9.42 gH₂ L⁻¹ but exhibiting poor gravimetric uptake (0.8-2.66 wt. %)³⁸. Consequently, achieving DOE targets at 77 K does not guarantee performance at DOE-recommended temperatures. In contrast, MOF-2087 exceeds the gravimetric storage of MOF-5, IRMOF-20, and PCN-610²¹ by 621.2%, 342.75%, and 186.8%, respectively, under pressure swing conditions at 298 K, showcasing exceptional; gravimetric hydrogen storage at ambient temperature. These performance differences stem from MOF-2087's enhanced porosity and surface areas. For instance, the modest surface area (2200-2500 m² g⁻¹), limited pore volume (0.4-1.3 cm³ g⁻¹), and simple Zn₄O secondary building unit of MOF-5 result in rapid drop in hydrogen uptake at elevated temperatures. In contrast, the larger unit cell (39 Å x 39 Å x 39 Å), higher pore volume (~5 cm³ g⁻¹) and presence of aromatic carbon clusters in MOF-2087 provide an efficient network for better hydrogen adsorption. Although MOF-2087's volumetric capacity reduces by 21.02%, 26.22%, and 24.2% compared with MOF-5, IRMOF-20, and PCN-610, there is still room to enhance volumetric uptake through strategies such as alkali metal decoration, monolith formation, or alternative operating scenarios like temperature-pressure swing conditions²³.

Fig. 14. Trade-off comparison of MOF-2087 with experimental and computational frameworks^{29,59,60}.

As expected, the gravimetric hydrogen storage of MOF-2087 (4.97 wt. %) is particularly noteworthy when compared to MOFs with dominant volumetric capacity (other than benchmarked MOFs, dotted red lines in Fig. 14), which often exhibit low gravimetric uptake. For example, Mg-MOF-74 and Zn-MOF-74 offer exceptional volumetric capacities of 12.3 gH₂ L⁻¹ and 11.96 gH₂ L⁻¹, respectively, but their gravimetric capacities are considerably lower at 0.91 wt. % and 0.67 wt. %, highlighting the tradeoff where enhanced volumetric capacity comes at the expense of gravimetric uptake (refer to Fig. 14). Although MOF-2087 has a modest volumetric uptake of 6.95 gH₂ L⁻¹, it achieves a superior gravimetric uptake, suggesting a favorable structure that delivers significant hydrogen uptake without compromising volumetric density.

Further comparison with NU-1501-Al, which has a near-similar topology and crystallographic features, reveals that MOF-2087, with its higher number of carbon atoms and 8-membered C-cluster-like porous aromatic frameworks⁶¹, exhibits greater hydrogen uptake²⁹. While both MOFs have similar volumetric capacities, the gravimetric capacity of MOF-2087 is double that of NU-1501-Al (Fig. 14). The metal centers and oxygen atoms in MOF-2087 contribute to hydrogen adsorption due to significant electronegativity differences, while the C-cluster traps hydrogen via van der Waals interactions, as indicated by the C_2 and C_R carbon interaction results from MD simulations. The presence of diverse, active sites, including the metal center, oxygen atoms, and C-cluster, as revealed by the mass center probability distribution (Fig. 6), likely accounts for the higher gravimetric capacity. At the same time, the slight deviation of crystallographic features of NU-1501-Al from the ideal solution, especially with void fraction and volumetric surface area, could have also been attributed to the reduced gravimetric uptake. As the temperature increases from 77 K to 298 K, a significant drop in hydrogen uptake is observed, similar to NU-1501-Al, which is attributed to the increased kinetic energy of hydrogen molecules.

Nonetheless, the uptake of MOF-2087 at 298 K and 100 bar (5.3 wt. % and 7.4 gH₂ L⁻¹) remains **notable**, demonstrating one of the best deliverable capacities with an **optimal tradeoff** under ambient conditions. **The observed performance at 298 K can be attributed to the structural features identified during optimization at 77 K, which remains stable across this temperature range. This was confirmed through simulated annealing, where the structure of MOF-2087 was relaxed at each temperature point, including 300 K. The framework remained stable throughout the range of 77-298 K, supporting the use of 77 K-optimized features for performance evaluation at elevated temperatures. Additionally, previous studies have shown that MOFs such as SNU-77 and Cd-doped variants retain their structural features from cryogenic to above ambient temperatures (373 K), further supporting this**

assumption^{62,63}. Therefore, MOF-2087 is expected to maintain the optimum trade-off in hydrogen storage performance even at ambient conditions.

Regarding comparative methods analysis, the B-RFT-MOPSO hybrid framework offers a significantly faster alternative to the brute-force high throughput GCMC screening process. This study's B-RFT framework required approximately 6-120 CPU seconds to calculate hydrogen storage densities for the global MOF features. The MOPSO algorithm then took around 168 CPU hours to conduct an average of 109,437 framework evaluations to identify the local best solutions and less than 10 CPU minutes to determine the global best features from the local repository. Although the B-RFT-MOPSO framework necessitates pre-training and validation datasets from experimental GCMC simulations, it is scalable to larger search domains encompassing a wide range of features. Once evaluated and validated, the framework enables computationally efficient evaluation of new feature combinations, potentially running on a desktop workstation rather than requiring HPC servers, as the current study employed an Intel Xeon 24-core machine. In contrast, brute-force GCMC screening is computationally expensive and intensive, often dependent on the number of equilibration and production cycles the user sets. For instance, Snurr et al. reported that screening a dataset of 130,000 hMOFs would require approximately 500,000 CPU hours, impractical for low-configuration machines⁶⁴. Employing the optimization-based screening approach centered on the statistical ideal solution; the same task could be accomplished in less than 200 CPU hours for a single run, effectively isolating feasible MOF feature combinations.

Hence, in a nutshell, the hydrogen sorption of MOF-2087 is undoubtedly superior and can fit into the leaderboard of benchmarked MOFs. This peculiar nature of MOF-2087 also highlights the reliability, search ability, and tendency to reach optimality of our optimization-based screening approach (B-RFT-MOPSO). In summary, these observations show our optimization-

based searching route as a reliable and the most promising route for material selection from a realm of available materials.

Conclusions

The present work demonstrates the B-RFT-MOPSO hybrid computational framework as a powerful approach for optimizing and screening MOFs for hydrogen storage. MOF-2087 was identified as the best-performing framework from a global repository of 733,792 MOFs, achieving an optimal tradeoff between gravimetric and volumetric hydrogen storage capacities, 38 g H₂ L⁻¹, and 22.3 wt. % at 77 K, 100 bar, and 5.3 wt. % and 7.4 gH₂ L⁻¹ at 298 K, 100 bar, outperforming classical benchmarked MOFs like MOF-5, IRMOF-20, PCN-610.

Simulated annealing and MD simulations revealed that in MOF-2087, C₂ and C_R atoms are the most favorable adsorption sites for hydrogen due to their π - π interactions and the electronegativity of sp² hybridized carbons. The radial distribution function showed a significant decrease in the distance between C₂ and H₂ from 5.9 Å at 300 K to 3.6 Å at 14 K, indicating enhanced adsorption at lower temperatures. Similarly, C_R-H₂ distances decreased, highlighting strong π - π stacking interactions. O atoms also demonstrated strong adsorption potential, with distances reducing from 6.2 Å to 4.2 Å, while Zn showed moderate interaction, likely due to polarization effects. H₂ diffusion analysis indicated increased mobility with temperature, but lower diffusion rates at reduced temperatures facilitated access to favorable adsorption sites. These findings, consistent with GCMC simulations, underscore the role of C₂ and C_R atoms as primary adsorption sites, contributing to the enhanced hydrogen uptake in MOF-2087.

In future, the proposed machine learning and multi-objective optimization framework can be extended to optimize functional group combinations and metal ion selections, enabling the design of new MOFs with enhanced

tradeoffs at 298 K. Furthermore, the proposed computational framework offers a scalable and versatile approach for identifying high-promising MOFs specifically tailored for onboard hydrogen storage in fuel cell vehicles. Beyond this primary focus, the method can also be adapted to other real-world gas storage applications such as methane by retraining the B-RFT model with domain-specific datasets.

4. Methods

4.1 Data Mining and Screening

A global repository of 925,229 MOFs was mined from the HyMARC (Hydrogen Materials Advanced Research Consortium) data hub, which complies with data from 19 known databases and was deposited by Ahmed et al.²¹. Initial screening classified 191,437 MOFs as non-open-source structures (i.e., their crystallographic information is not publicly available) and 733,792 MOFs as open-source structures. These open-source MOFs were shortlisted and stored in the labelled repository, the global database. **Supplementary Fig. 11** illustrates the distribution of MOFs across the global database based on their respective sources. The seven key crystallographic features, crystal density, gravimetric and volumetric surface areas, pore volume, pore diameter, cavity diameters, and void fractions, were identified using HyMARC data hub^{21,23,65}. A subset of the global database, consisting of MOFs with deliverable hydrogen storage capacities measured at 100-5 bar pressure swing at 77 K^{21,23}, was extracted to train the B-RFT framework (**Supplementary Table 5**). This training dataset includes hydrogen storage capacities of 12,764 experimental MOFs (eMOFs) and 85,930 hypothetical MOFs (hMOFs). The distribution of usable gravimetric and volumetric capacities within the subset is depicted in **Supplementary Fig. 12**.

Including both eMOFs and hMOFs in the training dataset significantly enhances the performance of the B-RFT in predicting hydrogen storage

densities across the global database. The dataset was shuffled before training and testing to prevent the B-RFT from memorizing trends in the training data. Correlation analysis of the training data revealed that gravimetric capacity is predominantly influenced by pore volume and gravimetric surface area. In contrast, volumetric capacity is mainly governed by a void fraction of MOFs (**Supplementary Fig. 2**). For simplicity, we refer to the repository of 733,792 open-source MOFs as the global database. Detailed steps involved in the present study are provided in **Supplementary Fig. 13**.

4.2 B-RFT Framework and Performance Evaluation

The B-RFT (Bootstrap Aggregated Random Forest Tree) framework couples randomization with decision trees, enabling efficient prediction for larger datasets⁶⁶. By implementing parallel decision trees trained on random subsets of data, the framework reduces the chances of overfitting and improves prediction accuracy for complex datasets. This adaptive bootstrapping strategy enhances scalability and efficiency in large-scale onboard applications. **Supplementary Fig. 14** outlines the hydrogen storage density prediction steps using the B-RFT framework, with additional details in **Supplementary Method 1**.

The B-RFT model aims to predict the usable gravimetric and volumetric hydrogen storage densities in MOFs under pressure swing conditions at 77 K. Separate B-RFT models were designed for each density to reduce the complexities, evaluated using performance metrics such as R^2 , RMSE, AUE, and MEA²¹. Model sensitivity analysis was performed on training dataset sizes ranging from 4,935 to 78,965 MOFs, with 9,870 MOFs used for validation and testing^{21,67}. The B-RFTs best-performing model was used as the fitness function for subsequent multi-objective particle swarm optimization (MOPSO) due to the lack of an explicit fitness function for optimizing hydrogen storage tradeoffs.

4.3 MOPSO

The MOPSO^{68,69} algorithm was employed to identify MOFs with optimal tradeoffs between hydrogen storage capacities. In multi-objective optimization, a local repository was created to store non-dominated solutions, updated during iterations. To ensure diversity of search, repository size was constrained, and the algorithm continued until convergence criteria were met. The B-RFT-MOPSO framework was configured to maximize MOF's hydrogen storage densities, with design parameters such as population size, learning coefficients, and damping ratios tailored from previous optimization studies^{39,70,71}. For additional details refer to **Supplementary Fig. 15** and **Supplementary Methodology 2**. Performance metrics including the standard spacing metric (SSM), standard generational distance (SGD), and solution set size were used to evaluate the framework's effectiveness. Further details of MOPSO, performance indicators, nearest neighbor search, and multi-criteria decision-making technique are provided in **Supplementary Methodologies 2 and 3**.

4.4 GCMC and MD approach

The hydrogen adsorption properties of potential MOFs identified through B-RFT-MOPSO were simulated using the GCMC method via the RASPA computational package⁵⁸. Adsorption isotherms for each pressure point were calculated over 10,000 cycles (5,000 for equilibration and 5,000 for data collection). In GCMC, each cycle corresponds to 'm' Monte Carlo moves, where 'm' is either 20 or the number of adsorbate molecules in the supercell, whichever is higher. Each simulation move includes translation, rotation, insertion, or deletion of guest molecules with equal probabilities. The intermolecular interactions between the host framework and adsorbate molecules were modeled using Lennard-Jones (LJ) potentials (Eq. 1). Long-range corrections were applied with a cutoff radius of 12 Å.

$$V_{ij}^{LJ} = 4\epsilon_{ij} \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right) \dots \dots \dots \text{Eq. (1)}$$

Here, ϵ_{ij} represents the depth of potential, r_{ij} is the distance between atoms i and j , and σ_{ij} corresponds to the distance at which the intermolecular potential between two particles equals zero. Cross-interactions between host and adsorbate molecules were calculated using the Lorentz-Berthelot missing rule:

$$\epsilon_{ij} = \sqrt{\epsilon_i \epsilon_j} \text{ and } \sigma_{ij} = \sqrt{\sigma_i \sigma_j} \dots\dots\dots \text{Eq.}$$

(2)

Hydrogen molecules were modeled using a single van der Waals (vdW) site, as Fischer et al.⁷² proposed. The adsorbate-adsorbate columbic interactions were computed using Ewald summation.

The adsorbate-host framework and adsorbate-adsorbate interactions were corrected using pseudo-Feynman-Hibbs correction^{24,73} to account for the quantum effects in low-temperature regimes, which adjusts the LJ potential based on temperatures:

$$V_{ij}^{\text{feynman-hibbs}} = V_{ij}^{\text{LJ}}(r) + \frac{\hbar^2}{24 \mu k_B T} \Delta^2 V_{ij}^{\text{LJ}}(r) \dots\dots\dots \text{Eq. (3)}$$

Where k_B is the Boltzmann constant, μ the reduced mass, $\hbar$ is the Planck constant, and T the temperature. Framework molecules were modeled using the universal force field⁷⁴ and DREIDING⁵³ for the metal and non-metal framework atoms respectively⁷⁵. The supercell structure was tailored for each MOF to ensure it was sufficiently large relative to twice the cutoff radius, with all MOFs treated as rigid.

The probability density distribution of hydrogen and radial distribution functions (RDFs) were calculated to identify preferred adsorption sites within the MOFs. The RDF between hydrogen and host atoms was calculated as follows⁵¹:

$$g(r) = \frac{\Delta N_{ij} V}{4\pi r^2 N_i N_j} \dots\dots\dots \text{Eq.}$$

(4)

Where V is the system volume, N_i and N_j are the number of host and guest atomic pairs, and r is the distance between the centers of mass of the pairs. The isosteric heat of adsorption (Q_{st}), representing the interaction energy between host and guest molecules, was calculated based on potential energy (ν) and the number of adsorbate molecules (N) as shown in Eq. 5:

$$Q_{st} = RT - \left(\frac{\partial \langle \nu \rangle}{\partial \langle N \rangle} \right)_T \dots \dots \dots \text{Eq.}$$

(5)

Considering the ensemble fluctuations, Q_{st} can also be expressed as ^{76,77},

$$Q_{st} = RT - \frac{\langle \nu N \rangle - \langle \nu \rangle \langle N \rangle}{\langle N^2 \rangle - \langle N \rangle^2} \dots \dots \dots \text{Eq.}$$

(6)

The GCMC approach was validated by calculating the hydrogen storage capacities of 10 randomly selected MOFs from the training dataset²¹.

Supplementary Fig. 15 shows a close agreement between the training dataset and present GCMC computed capacities, confirming the model's reliability.

Classical molecular dynamics simulations were performed in LAMMPS package, to understand the global best MOF's hydrogen storage mechanism. Similar force fields and hydrogen model used in GCMC were applied. MD simulated annealing was carried out to identify the minimum potential energy sites in the MOF. The number of hydrogen molecules in the simulation cell was based on GCMC, adsorption data. Hydrogen molecules were randomly introduced into the cell, equilibrated at 300 K for 10,000 steps using NVT ensemble, and annealed from 300 K to 1 K over 10.5 ps. A longer MD simulation lasting 2,000 ps, was performed at various temperatures (77 K, 160 K, 200 K, and 298 K) to compute the mean square displacement (MSD)⁵⁴:

$$\text{MSD} = \frac{1}{N} \sum_{i=1}^d |R(t) - R_t(0)|^2 \dots \dots \dots \text{Eq.}$$

(7)

Where $R(t)$ is the center of mass of hydrogen molecule at time t , and N is the number of hydrogen molecules in the system. The diffusion coefficients (D_H) were then calculated using Einstein's relation⁷⁸.

$$D_H = \frac{1}{2n_{\text{dim}}} \int_{-\infty}^{\infty} \frac{1}{\tau} \langle \text{MSD} \rangle$$

.....Eq. (8)

Where n_{dim} is the diffusion dimensionality (3 for present system). The number of hydrogen molecules around the host atoms at increased radial distances was computed by using RDF $g(r)$.

$$N(r) = 4\pi \int_0^r r'^2 p g(r') dr' \dots \dots \dots \text{Eq. (9)}$$

Where p is the average hydrogen density around the host atoms.

Data Availability

- All the relevant information and necessary datasets were available in the Supplementary information file. The Supplementary Excel file showed the Pareto optimal solutions and local best MOFs. Further details or data will be available from the corresponding author upon reasonable request.
- The CIF information of MOF-2087 can be found in the Supplementary information file.
- The pretrained B-RFT models can be found in figshare public repository: <https://doi.org/10.6084/m9.figshare.28925930>

Author Contributions

A.P: Conceptualization, methodology, validation, data curation, formal analysis, visualization, writing-original draft. R.D: data curation, formal analysis, visualization, writing-original draft. S.R.: conceptualization, supervision. A.A.: supervision, funding acquisition. R.S.: supervision, funding

acquisition. All authors contributed to manuscript review and editing and have approved the final version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the presented work.

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